

STAT 7500 – Statistical Programming

Homework #2

Submission instructions: Please choose one of the following options to turn in your homework:

- 1) **A single file**, in Word or some other word processing program, that includes relevant information copied from the SAS editor(s) and the SAS output/reviewer file(s). In addition, please make sure to include your responses to specific inquiries in the assignment. For this assignment, this refers to the line in problem #1b that asks you to “Comment on what these outputs indicate to you”. Please also include your entire SAS (editor window) file as an Appendix. **Note - this is the preferred method of submitting this homework.**
- 2) **Multiple files** that include your SAS editor(s), the SAS output/reviewer file(s), and a place where you write responses to specific inquiries in the assignment. For this assignment, this refers to the line in problem #1b that asks you to “Comment on what these outputs indicate to you”. This could be either in a separate document that clearly identifies which problem you are responding to (i.e., #2 for this assignment) or embedded in the appropriate places as comments in your SAS editor(s). Note, Blackboard should accept multiple files from you, but if there is a problem, you should try zipping your files together and submitting a single zipped file.

Problem 1

Data were gathered on twenty patients at a clinic. The patients were given identification numbers, 001 – 020. A health awareness survey was given to the patients at baseline, which is scored out of a possible 20 points, with higher numbers indicating more awareness of health habits. These data are included in the file *Patients Time 1*. There should be no strange characters in the file – but I believe students have sometimes downloaded them in strange ways that created those characters. If you see any, let me know and I will either post a new version or send you the files directly.

- a) Read the file *Patients Time 1* into SAS as a SAS dataset using the following methods:
 - i. Copy and paste data into the SAS editor and read it in directly through the SAS editor. *Hint: ask Google or ChatGPT how to get datalines to read in raw data separated by commas.*
 - ii. Read in the external file using an PROC IMPORT statement.
- b) Run a PROC CONTENTS and a PROC PRINT on the dataset to make sure the data was read in properly. Comment on what these outputs indicate to you.
- c) Write *Patients Time 1* as a permanent SAS dataset.

Problem 2

- a) Read the file Patients Gender Data (which should be a tab delimited file) into SAS
- b) Apply the following attributes to the dataset:
 - i. Make the *gender* variable say “Female” when the value is 1 and “Male” if 0
 - ii. Make SAS print the title “Patient ID” instead of just “ID” for the *ID* variable.
- c) Run a PROC FREQ and comment on the proportion of males vs. females in the dataset.

Problem 3

Recall the class_scores data you worked with in SAS HW 01

```
data class_scores;
  input name $ & 20. test1 test2 test3 final;
  datalines;
  Alexandra May 80 85 75 90
  Greta Carlsen 85 85 90 90
  Harry Romano 75 85 95 85
  Marshall Wasserstein 100 90 95 100
;
run;
proc print data=class_scores noobs label;
  title "Student Class Test Scores";
  label test1="Test 1" test2="Test 2" test3="Test 3"
        final="Final Test";
run;
```

- a. Adapt the code so that it creates *permanent* labels for this data. Use PROC CONTENTS to show these permanent labels. Comment on why the labels created HW 01 were temporary.
- b. Create a temporary library named “homework” that points to a folder on SAS OnDemand. Save the class_scores data set to that library location under the name “class_scores_perm”. Confirm that the class_scores_perm data set is in two locations: (1) your temporary homework library and (2) the permanent location/folder that you pointed to when defining the homework library.
- c. Load in the class_scores_perm data set from the library location. Print out the dataset to confirm it matches the class_scores data set.

Problem 4

- a. Use PROC IMPORT to import the Babynames file from the course website. Inspect the data and comment on what it contains.
- b. Run a PROC FREQ and comment on what it tells you about the data.
- c. Create a new dataset that keeps only the years prior to 1900
- d. Run a PROC CONTENTS on the dataset you created in 4c